Predicting and Explaining Infant Mortality Rates in India Using District-Level Health Indicators and Machine Learning

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Interim Report

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QM640: Data Analytics Capstone

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Interim Report

Project Repository

GitHub Repository (Data and Code):

https://github.com/Sheeba-analyst/infant-mortality-rate-capstone

1.Introduction

Infant mortality is widely recognized as one of the most important indicators of a nation's public health status and socioeconomic development. The infant mortality rate (IMR), defined as the number of infant deaths before the age of one year per 1,000 live births, reflects the combined effects of maternal health, healthcare access, nutrition, sanitation, and broader socioeconomic conditions. Because infant survival is closely linked to the quality of prenatal and postnatal care, immunization coverage, and disease prevention, IMR is commonly used by governments and international organizations as a key measure of healthcare system performance and population well-being (Subramanian et al., 2024; World Health Organization, 2023).

Over the past two decades, India has made considerable progress in reducing infant mortality through improvements in maternal and child health programs, expanded vaccination coverage, and increased access to healthcare services. National statistics show a steady decline in infant mortality rates across the country. However, significant disparities still exist across states and districts due to differences in healthcare infrastructure, socioeconomic conditions, maternal health services, and disease burden. These regional variations highlight the need for more detailed district-level analysis to better understand the determinants of infant mortality and support targeted public health interventions (Suresh, 2025; Registrar General of India, 2022).

The growing availability of open government datasets has created new opportunities for applying advanced data analytics methods to public health problems. Platforms such as the National Data and Analytics Platform (NDAP) provide district-level data on maternal healthcare, immunization coverage, demographic indicators, and disease prevalence. Such datasets contain a large number of variables that may influence infant mortality outcomes. Machine learning techniques are particularly useful for analyzing high-dimensional datasets because they can identify complex relationships among variables and detect the most important predictors influencing health outcomes (Saroj et al., 2022).

Despite the availability of detailed health data, understanding how multiple healthcare and socioeconomic indicators jointly influence infant mortality remains challenging. Many traditional statistical approaches are limited when dealing with large numbers of predictors and complex interactions. As a result, identifying the most influential determinants of infant mortality at the district level requires more advanced analytical approaches. Machine learning models can help address this challenge by enabling predictive analysis and identifying key variables that contribute most strongly to variations in infant mortality across regions (Shaw et al., 2023; Vishwakarma et al., 2024).

The purpose of this study is to analyze district-level health and demographic indicators across India in order to identify the key determinants of infant mortality and develop predictive models using machine learning techniques. By combining exploratory data analysis, feature selection methods, and predictive modeling, this study aims to understand how factors such as maternal healthcare access, immunization coverage, disease burden, and demographic characteristics influence infant mortality outcomes. The insights generated from this analysis may support evidence-based public health planning and contribute to improved strategies for reducing infant mortality in high-risk districts.

2. Scope and Objectives

The scope of this project is to analyze district-level healthcare and socioeconomic indicators in India to understand the determinants of infant mortality and develop predictive models capable of estimating infant mortality risk across districts.

Research Questions:

RQ1: How do infant mortality and child health outcomes vary across districts in India?

RQ2: Which maternal healthcare, vaccination, disease, and socioeconomic indicators significantly influence infant mortality?

RQ3: Can machine learning models predict district-level infant mortality using healthcare indicators?

RQ4: How can an explainable AI interface help interpret machine learning predictions for policymakers?

3. Literature Survey

Infant mortality has been widely studied as a critical indicator of population health, reflecting the combined effects of healthcare access, maternal well-being, and socioeconomic conditions. Existing literature highlights that infant mortality is not driven by a single factor but by a complex interaction of healthcare, demographic, and economic variables.

Subramanian et al. (2024) conducted a comprehensive analysis of neonatal and child mortality trends across India, revealing substantial regional disparities in mortality outcomes. Their findings emphasize that national-level improvements often conceal localized inequalities, particularly across districts with varying healthcare infrastructure and socioeconomic development. **This study directly informs RQ1 by establishing the importance of examining district-level variation, which forms the basis of the current analysis. The key takeaway is that disaggregated data is essential for identifying high-risk regions and designing targeted interventions.**

Suresh (2025) further explored state-level differences in infant mortality and identified maternal healthcare access, immunization coverage, and institutional delivery as key determinants of improved child survival outcomes. The study highlights that preventive healthcare services play a critical role in reducing infant mortality. **This insight supports RQ2 and guides the selection of maternal and child health indicators in this study. The key takeaway is that strengthening maternal healthcare systems and vaccination programs is central to reducing infant mortality rates.**

From a methodological perspective, Saroj et al. (2022) demonstrated the effectiveness of machine learning algorithms in identifying determinants of under-five mortality using large-scale health datasets. Their work shows that machine learning models can capture nonlinear relationships and interactions that are often missed by traditional statistical methods. **This provides a strong foundation for RQ3, justifying**

the use of ensemble learning techniques such as Random Forest and Gradient Boosting in the current study. The key takeaway is that machine learning enhances predictive accuracy and variable discovery in high-dimensional health data.

Shaw et al. (2023) introduced explainable artificial intelligence (XAI) approaches for infant mortality prediction, emphasizing that predictive performance alone is insufficient in healthcare applications without interpretability. Their work highlights the importance of generating human-understandable explanations alongside model predictions to support decision-making. **This directly informs RQ4 and motivates the implementation of a simplified Retrieval-Augmented Generation (RAG) framework in this study. The key takeaway is that transparency and interpretability are essential for translating predictive analytics into actionable policy insights.**

Singh et al. (2025) applied time-series forecasting techniques to analyze trends in infant mortality over time. While their focus is on temporal dynamics rather than cross-sectional analysis, their findings demonstrate the value of predictive modelling in anticipating future health outcomes. **This complements the current study by reinforcing the role of predictive analytics in public health planning. The key takeaway is that forecasting models can support proactive and preventive healthcare strategies.**

Vishwakarma et al. (2025) utilized survival-based machine learning methods to model child mortality risk, capturing time-to-event dynamics and complex relationships between predictors. Their study demonstrates the robustness of advanced machine learning approaches in analyzing mortality outcomes. **This further supports the**

methodological framework of the current study under RQ3. The key takeaway is that advanced ML techniques can effectively model complex healthcare relationships and improve predictive insights.

Beyond individual studies, broader public health research consistently identifies antenatal care, maternal nutrition, institutional delivery, and immunization coverage as critical determinants of infant survival. These factors reflect both healthcare access and quality, as well as underlying socioeconomic conditions. **These insights guided the selection and categorization of variables in the present dataset, ensuring alignment with established public health frameworks.**

Recent developments in health analytics highlight the effectiveness of ensemble learning methods, particularly Random Forest and Gradient Boosting, in handling high-dimensional data and improving predictive performance. **This supports the modelling approach adopted in this study and aligns with RQ3.**

Furthermore, emerging research in health informatics emphasizes the integration of predictive models with decision-support systems to enhance policy implementation. Explainable AI techniques, including RAG-based frameworks, are increasingly being used to bridge the gap between complex models and end users such as policymakers. **This aligns with the objective of this study to develop an interpretable prediction system under RQ4.**

4. Data Description

Data Sources

This project uses publicly available datasets obtained primarily from the **National Data and Analytics Platform (NDAP)** developed by the NITI Aayog. NDAP provides integrated access to multiple government datasets across sectors such as health, demographics, and economic development, enabling large-scale data analysis for policy research.

The study integrates multiple district-level datasets covering health, socioeconomic, demographic, and economic indicators. The primary datasets used in this study include:

• **District Health Indicators Dataset** – This dataset contains **district-level** healthcare variables such as maternal healthcare indicators, immunization coverage, disease prevalence, infant deaths, and birth-related statistics. These indicators provide key information related to maternal and child health conditions that may influence infant mortality outcomes.

• **Socioeconomic Indicators Dataset** – This dataset includes **district-level** socioeconomic variables such as sanitation coverage, literacy levels, and household characteristics. These indicators help capture broader social determinants of health that may indirectly affect infant survival and healthcare access.

• **MSME Indicators Dataset** – This dataset contains **district-level** information related to **Micro, Small, and Medium Enterprises (MSMEs)** obtained from the Ministry of Micro, Small and Medium Enterprises. MSMEs are classified based on factors such as investment, size, and employment levels. The dataset also includes gender-wise classifications of enterprises based on ownership or management. Such indicators provide insights into economic activity, employment opportunities, and regional economic development, which may influence household income levels and access to healthcare resources.

All datasets were merged at the **district level** using **district names as the primary key after performing preprocessing steps** such as cleaning, standardizing district names, and removing inconsistencies across datasets. The integrated dataset provides a comprehensive set of indicators that capture health, demographic, socioeconomic, and economic characteristics of districts across India, enabling a data-driven analysis of the factors associated with infant mortality.

After merging datasets based on common district identifiers, 553 districts with 314 indicators were initially retained for analysis, which were subsequently reduced to 33 predictors through feature selection.

5. EDA and Analysis

5.1. Data Cleaning and Preprocessing

The initial dataset was created by merging multiple district-level datasets using common district identifiers. Only districts that were present across all datasets were retained to ensure consistency in the analysis. After merging, the dataset initially contained **317 variables**.

Several **metadata and identifier columns** that were not relevant for analysis were removed. These included administrative identifiers such as state codes, district codes, and calendar-related fields. However, **state and district names were retained**

separately for reporting purposes. Removing these metadata variables reduced the number of features from **317 to 310 columns**.

Data Integration and Alignment

The datasets used in this study originate from different sources and collection frameworks, including HMIS and NFHS. These datasets may differ in reporting periods and administrative boundary definitions.

To address this, the data were aligned at the district level using common district names. Only districts consistently present across all datasets were retained for analysis. While minor inconsistencies may remain due to temporal and boundary variations, this approach ensures a harmonized dataset suitable for exploratory and predictive analysis. This limitation is acknowledged and considered when interpreting results.

5.2. Missing Value Treatment

The dataset was then examined for missing values. Variables with a high proportion of missing data may reduce model reliability and introduce bias. Therefore, columns with **more than 30% missing values were removed**, which reduced the number of variables from **310 to 295**.

For the remaining variables, missing values were handled using **median imputation**. Median imputation was chosen because it is robust to outliers and helps preserve the overall distribution of the data.

5.3 Feature Reduction

To address the large number of variables and potential multicollinearity, a **correlation-based feature reduction** method was applied. The absolute correlation matrix was computed and variables with pairwise correlations greater than **0.80** were considered highly redundant. One variable from each highly correlated pair was removed, reducing the number of variables from **295 to 215**.

Since the study focuses on **infant mortality (deaths under one year of age)**, several variables related to **children older than one year** were removed to maintain conceptual relevance to the research objective. Columns containing age group patterns such as **"12 to 23 months," "under age 3," "under 5," "6 to 59 months," "9 to 35 months," and "age 5 years"** were excluded. After removing these variables, the dataset contained **189 features across 562 districts**.

5.4 Target Variable Construction

The target variable for the analysis, **infant mortality rate**, was computed using reported infant deaths and live births available in the dataset. The infant mortality rate (IMR) was calculated as:

$$IMR = \frac{\text{Infant deaths reported}}{\text{Reported live births}} \times 1000$$

This variable represents the number of infant deaths per **1,000 live births** and serves as the dependent variable in the predictive modelling stage.

5.5 Feature Scaling

Before applying machine learning models, the feature variables were standardized using **standard scaling**. Standardization ensures that variables measured on different scales contribute equally to the model by transforming them to have a **mean of zero and standard deviation of one**. This step is particularly important for models that are sensitive to feature scaling, such as regularized regression models.

5.6 Data Leakage Prevention

To avoid **data leakage and circular prediction**, variables directly related to the calculation of the infant mortality rate (such as **infant deaths and reported live births**) were removed from the predictor dataset before model training. This ensures that the machine learning models do not use variables that directly determine the target variable, which would otherwise lead to artificially inflated model performance.

5.6 LASSO Feature Selection

To further reduce dimensionality and identify the most important predictors, **LASSO (Least Absolute Shrinkage and Selection Operator) regression** with cross-validation was applied for feature selection. LASSO performs both regularization and variable selection by shrinking the coefficients of less important variables toward zero.

Using **5-fold cross-validation**, LASSO automatically determined the optimal regularization parameter and selected variables whose coefficients remained non-zero. After applying LASSO feature selection, the number of predictors was reduced significantly, and **33 variables were retained** as the most relevant features for predicting infant mortality.

This final reduced dataset was then used for subsequent modelling and analysis to understand the determinants of infant mortality across districts.

5.7 Descriptive and Exploratory Data Analysis

Before conducting inferential statistical tests, a comprehensive descriptive and exploratory analysis was performed to understand the distribution and determinants of the computed infant mortality rate (IMR) across districts. Summary statistics, including the minimum, maximum, mean, and standard deviation, were calculated to assess the variability and range of IMR values. The analysis revealed considerable variation in infant mortality across districts, indicating substantial disparities in child health outcomes.

To further examine these differences, districts were ranked based on IMR values, and the ten districts with the lowest and highest IMR were identified. This comparison provides a clear contrast between regions with relatively better child health outcomes and those experiencing higher levels of infant mortality, offering initial insights into geographic inequalities.

In addition to IMR-specific analysis, exploratory data analysis of key maternal health, nutrition, and healthcare access indicators was conducted. The results highlight significant heterogeneity across districts. While infrastructure-related variables such as electricity access show consistently high coverage with limited variability, critical health indicators—including maternal anaemia, institutional births, and iron and folic acid consumption—exhibit wide dispersion, reflecting disparities in healthcare access and utilization. Several variables also display skewed distributions and the presence of outliers, often influenced by population differences across districts, indicating the need for careful interpretation and preprocessing.

Bivariate analysis further suggests that maternal and healthcare-related variables demonstrate stronger relationships with IMR compared to general infrastructure indicators, although these relationships are often non-linear and influenced by multiple interacting factors. Overall, the exploratory analysis underscores that infant mortality is a multidimensional issue shaped by both healthcare and socioeconomic inequalities across districts.

The detailed exploratory analysis, including descriptive statistics, distribution plots, box plots, and correlation visualizations, is provided in **Appendix A.**

5.8 Minimum Sample Size Computation

Although this study uses district-level population data, minimum sample size requirements were assessed for each research question to ensure methodological validity.

Table1: Minimum Sample Size Computation

Research Question	**Method Used**	**Approach for Sample Size**	**Minimum Required Sample Size**
RQ1	Descriptive Analysis + ANOVA	Rule-of-thumb: ≥20 observations per group (Region as grouping variable, 6 regions; all groups >20)	≥120

RQ2	Correlation + Multiple Linear Regression	Green's Rule: $n \geq 50 + 8k$ ($k = 33$ predictors)	314
RQ3	Machine Learning Models (RF, GB, XGB)	Empirical ML requirement: large dataset with train-test split (80/20)	≥500
RQ4	Explainable AI (RAG-based interpretation)	Requires sufficient observations for stable prediction + explanation	≥500

The final dataset contains 558 district-level observations, which satisfies all minimum sample size requirements for statistical analysis, regression modelling, and machine learning approaches.

Note: ANOVA is conducted using broader geographical regions (North, South, East, West, Central, and Northeast) as the grouping variable to ensure sufficient observations per group. Although analysis using *State* as the grouping variable was also performed, some states had fewer than 20 districts. Therefore, region-based grouping was used to improve the statistical reliability and robustness of the results.

5.9 ANOVA Analysis

To evaluate differences in infant mortality, a one-way Analysis of Variance (ANOVA) was conducted using district-level IMR as the dependent variable. A regional grouping (North, South, East, West, Central, and Northeast) was first employed to ensure adequate observations within each group and to enable more balanced comparisons. The **results revealed a statistically significant difference in IMR across regions, F (5, 552) = 3.71, p = 0.0026.** This indicates that **geographical region is an important determinant of infant mortality**, reflecting underlying variations in socio-economic conditions, healthcare infrastructure, and access to maternal and child health services across different parts of India.

To complement this analysis, ANOVA was also conducted using *State* as the grouping variable. The findings similarly showed statistically significant variation in IMR across states ($p < 0.05$), reinforcing the presence of spatial disparities in infant mortality outcomes. However, since some states have relatively fewer districts, the region-based grouping provides a more statistically robust framework by ensuring sufficient sample size within each category while still preserving meaningful geographic distinctions.

Overall, the results from both state-level and region-level analyses consistently highlight the existence of significant disparities in infant mortality, emphasizing the importance of geographically targeted public health interventions.

5.10 Correlation Analysis

Further exploratory analysis examined the relationships between infant mortality and various health-related indicators. Correlation analysis revealed that variables such as maternal anaemia prevalence, low birth weight among newborns, and disease burden indicators tend to show positive associations with infant mortality, suggesting that higher prevalence of these conditions is linked with higher infant mortality levels.

In contrast, indicators related to antenatal care coverage, postpartum health checkups, and immunization coverage demonstrate negative correlations with infant mortality. These results suggest that improved maternal healthcare services and preventive health interventions may have a protective effect in reducing infant mortality rates.

Overall, these descriptive and exploratory analyses provide important preliminary insights into the distribution and potential determinants of infant mortality across districts, guiding the subsequent machine learning modelling phase of the study.

6. Modelling

To identify determinants of Infant Mortality Rate (IMR) and develop predictive models, both **statistical modelling and machine learning techniques** were applied. These approaches address **RQ2–RQ4** by examining relationships between healthcare indicators and IMR, predicting district-level mortality rates, and improving interpretability of model predictions.

6.1 Feature Selection

A multi-stage feature selection pipeline was used to reduce the dimensionality of the dataset and identify key predictors. Correlation filtering first reduced the variables from **208 to 120**, followed by **Mutual Information filtering** which retained **60 variables** with stronger relationships to IMR. Finally, **LASSO regression** was applied to select approximately **25 key predictors** used in modelling.

The selected variables represent major public health dimensions including **maternal healthcare, immunization coverage, disease burden, healthcare system utilization, and socioeconomic indicators**.

6.2 Multiple Linear Regression (RQ2)

A **Multiple Linear Regression (MLR)** model was developed to examine the statistical relationship between healthcare indicators and infant mortality.

The model produced the following results:

- **R^2:** 0.574
- **Adjusted R^2:** 0.548
- **F-statistic:** 21.58
- **Model significance:** $p < 0.001$

These results indicate that the selected predictors explain approximately **55% of the variation in district-level infant mortality rates**, suggesting a strong relationship between health indicators and IMR.

Several variables were found to be statistically significant predictors of infant mortality, including indicators related to **antenatal care registration, maternal nutrition (iron–folic acid consumption), hygienic menstrual practices, household electricity access, health insurance coverage, and healthcare utilization indicators such as inpatient deaths and maternal death reviews**. These findings highlight the importance of maternal health services, healthcare accessibility, and socio-economic conditions in influencing infant survival outcomes.

6.3 Multicollinearity Assessment

To ensure reliability of the regression estimates, **Variance Inflation Factor (VIF)** analysis was conducted to detect multicollinearity among predictors.

All variables had **VIF values below 10**, with most predictors ranging between **1 and 4**, indicating that multicollinearity is not a major concern in the model. This suggests that the regression coefficients are stable and suitable for interpretation.

6.4 Residual Diagnostics

Residual analysis was performed to evaluate whether the assumptions of linear regression were satisfied. The residual plot showed that the residuals were **randomly distributed around zero**, indicating that the assumption of linearity is reasonably satisfied.

The variance of residuals remained relatively consistent across predicted values, suggesting **no strong evidence of heteroscedasticity**. A small number of observations exhibited larger residuals, representing districts with unusually high infant mortality rates. These observations were retained as they reflect real-world variations in healthcare outcomes.

6.5 Machine Learning Models (RQ3)

To improve predictive performance, several **machine learning models** were implemented to predict district-level IMR.

The models evaluated include:

- **Random Forest**

- **Gradient Boosting**
- **XGBoost**

The dataset was divided into **training and testing sets using an 80:20 split**. Model performance was evaluated using standard regression metrics including **R^2, Root Mean Squared Error (RMSE), and Mean Absolute Error (MAE)**.

6.6 Explainable AI Approach (RQ4)

Although machine learning models can achieve strong predictive performance, their outputs are often difficult to interpret. To address this limitation, a **simplified Retrieval-Augmented Generation (RAG) approach** was implemented to provide explanations alongside model predictions.

For any selected district, the system retrieves key feature values and identifies the most influential predictors used by the model. These factors are then used to generate **human-readable explanations describing the drivers of predicted infant mortality rates**.

This approach improves transparency and helps translate predictive results into insights that can support **public health decision-making and policy planning**.

Evaluation of RAG Framework

The RAG-based explanation system will be evaluated qualitatively by comparing generated explanations with statistical model outputs and known domain relationships.

The explanations will be assessed based on:

- Consistency with regression coefficients and feature importance
- Interpretability and clarity for decision-making
- Alignment with established public health knowledge

While no formal quantitative evaluation metric is used, the framework demonstrates practical usefulness as a decision-support tool. Future work can incorporate expert validation or structured evaluation benchmarks to further strengthen this component.

The current RAG-based explanation framework represents a preliminary prototype designed to demonstrate the feasibility of integrating model predictions with natural language explanations. Further enhancements, including improved retrieval mechanisms, structured evaluation metrics, and user interaction capabilities, is planned to be developed and refined in the final report.

7. Preliminary Results

7.1 Multiple Linear Regression Results (RQ2)

The regression analysis demonstrated that district-level infant mortality rates are significantly associated with multiple healthcare and socio-economic indicators. The model achieved an **R^2 of 0.574 and an adjusted R^2 of 0.548**, indicating that more than half of the variation in IMR can be explained by the selected predictors.

Statistically significant variables ($p < 0.05$) included:

- Reported live births (to reported births)

- Women using hygienic menstrual protection methods
- Facility-based maternal death reviews
- In-patient department (IPD) deaths
- Antenatal care (ANC) registration
- Iron–folic acid consumption during pregnancy
- Health insurance coverage
- Blood transfusion treatment for complicated pregnancies
- Population sex ratio
- Household electricity access
- Albendazole treatment during pregnancy

Several of these variables are related to **maternal healthcare and healthcare system utilization**, reinforcing the importance of maternal health services in improving infant survival outcomes

Table1: Key Determinants

Category	**Variables**
Maternal Healthcare	ANC registration, Iron–Folic Acid
Healthcare Utilization	IPD deaths, maternal death review
Socioeconomic	Electricity access, insurance

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7.2 Machine Learning Model Performance (RQ3)

Machine learning models were developed to predict infant mortality rates using the selected predictors. The performance of the models was evaluated using **R^2, RMSE, and MAE** metrics.

Among the models tested, **Gradient Boosting achieved the best predictive performance**, with an **R^2 of approximately 0.516 and the lowest prediction error**. Random Forest also performed comparably, while **XGBoost showed slightly lower predictive performance under the default hyperparameter settings.**

Overall, the results demonstrate that machine learning techniques can effectively predict district-level infant mortality patterns using healthcare indicators

Table 2: Model Performance Comparison

Model	**R^2**	**RMSE**	**MAE**
Gradient Boosting	0.516	6.812	5.123
Random Forest	0.509	6.860	5.086
XGBoost	0.497	6.943	5.084

7.3 Comparison of Regression and Machine Learning Results

Interestingly, the predictive performance of the machine learning models was **similar to the multiple linear regression model**. The regression model achieved an adjusted R^2 of approximately **0.55**, which is comparable to the performance of Gradient Boosting.

This suggests that the relationships between healthcare indicators and infant mortality are **largely linear**, and traditional statistical models are capable of capturing much of the underlying structure in the data. As a result, regression models remain valuable because they provide **greater interpretability while maintaining competitive predictive performance**.

7.4 Explainability Insights using RAG (RQ4)

The simplified RAG-based explanation framework was used to interpret model predictions by highlighting the most influential factors contributing to predicted IMR values.

The generated explanations indicate that districts with higher predicted IMR tend to exhibit:

- Lower antenatal care coverage
- Lower immunization levels
- Higher prevalence of maternal health risks
- Increased healthcare burden reflected in hospital admissions and deaths

Conversely, districts with lower predicted IMR generally demonstrate stronger maternal healthcare systems, improved child nutrition, and better access to healthcare services.

This explainable framework helps translate model predictions into meaningful insights for policymakers and healthcare administrators.

7.5 Overall Evaluation

The modelling results demonstrate that both **statistical regression and machine learning methods provide valuable insights into infant mortality determinants**. Maternal healthcare indicators, healthcare system utilization, and socioeconomic conditions consistently emerge as key predictors across models.

While machine learning models provide strong predictive capabilities, the similar performance of regression models highlights the importance of **interpretable models in public health research**.

Overall, the preliminary results indicate that data-driven modelling approaches can effectively support **identification of high-risk districts and inform targeted healthcare interventions aimed at reducing infant mortality**.

8. Bibliography

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Appendix A

EDA Results and insights

Table1: Descriptive Statistics

index	Mothers who consumed iron folic acid for 100 days or more when they were pregnant (%)	Women registered for Ante Natal Care (ANC)	Population living in households with electricity (%)	Pregnant women age group 15 to 49 years who are anaemic (%)	Institutional births in public facility (%)
count	561	561	561	561	561
mean	45.80909	42260.82	96.96221	46.13993	65.18556
std	21.73338	37623.69	4.267188	19.15991	16.32258
min	2.6	150	68.4	0	18.1
25%	26.5	15866.33	96.3	37.9	53
50%	46.5	31870.67	98.6	49.6	66.8
75%	61.3	58732	99.5	59.4	76.9
max	95	231841	100	84.4	96.7

1. Descriptive Statistics Insights

Iron Folic Acid Consumption

Mean ≈ 45.8%, median ≈ 46.5% → well-centered distribution

Wide range (2.6% → 95%) → large inter-district variation

Insight:

The proportion of mothers consuming iron and folic acid is relatively balanced around the mean but shows substantial variation across districts. This indicates uneven access to maternal nutrition programs and healthcare services, which may contribute to disparities in infant health outcomes.

Antenatal Care (ANC Registration)

Mean ≈ 42,260 with very high standard deviation (≈ 37,623)

Range: 150 → 231,841 → highly skewed

Insight:

ANC registration exhibits extreme variability primarily driven by differences in district population size. The presence of very high values in certain districts suggests that this variable reflects scale rather than efficiency or quality of healthcare services, and should be interpreted cautiously in analysis.

Electricity Access

Mean ≈ 96.96%, median ≈ 98.6%

Very low standard deviation (≈ 4.26)

Insight:

Access to electricity is consistently high across districts, with minimal variation. While this indicates widespread infrastructure availability, the lack of variability limits its usefulness in explaining differences in infant mortality outcomes.

Maternal Anaemia

Mean ≈ 46.1%, median ≈ 49.6%

Wide range (0% → 84.4%)

Insight:

Maternal anaemia shows high prevalence and considerable variation across districts. This highlights it as a critical public health issue and a potentially strong determinant of infant mortality, reflecting disparities in maternal nutrition and healthcare access.

Institutional Births

Mean ≈ 65.2%, range (18% → 96.7%)

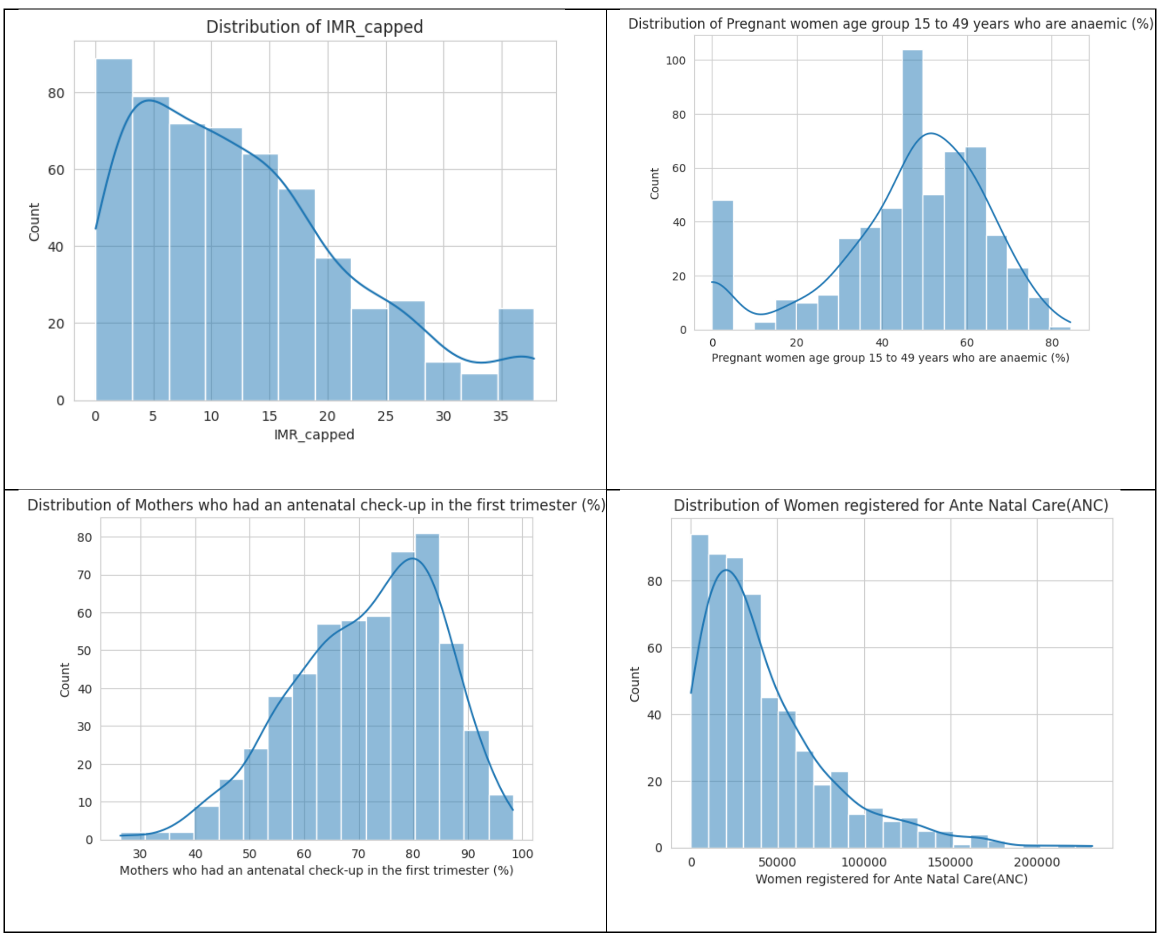
Insight:

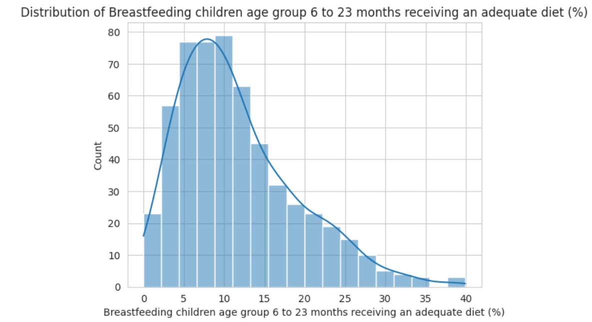
Institutional delivery rates vary widely, indicating differences in access to healthcare facilities and medical support during childbirth. Lower institutional birth rates in some districts may be associated with higher risks of complications and increased infant mortality.

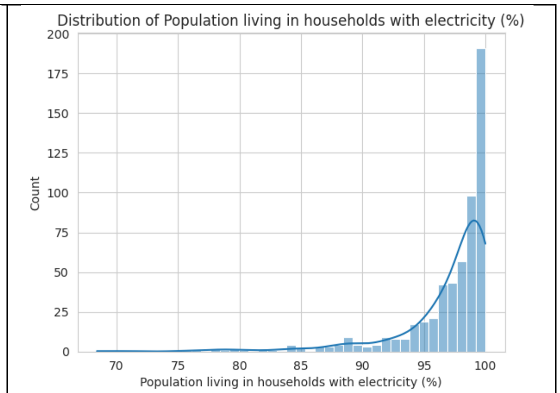
Summary

The descriptive statistics reveal substantial heterogeneity across districts in maternal health, healthcare utilization, and nutrition-related indicators. While infrastructure variables such as electricity access show uniformity, key maternal and healthcare indicators exhibit wide variation. This suggests that disparities in healthcare access and maternal conditions are likely key drivers of differences in infant mortality across districts.

Table2: Univariate Analysis- Distribution plots:







2. Univariate Analysis – Distribution Plots

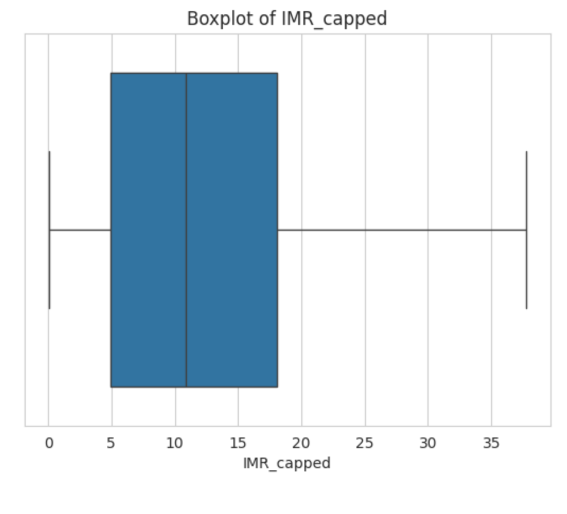
Most variables exhibit **non-normal and skewed distributions**, indicating heterogeneity across districts.

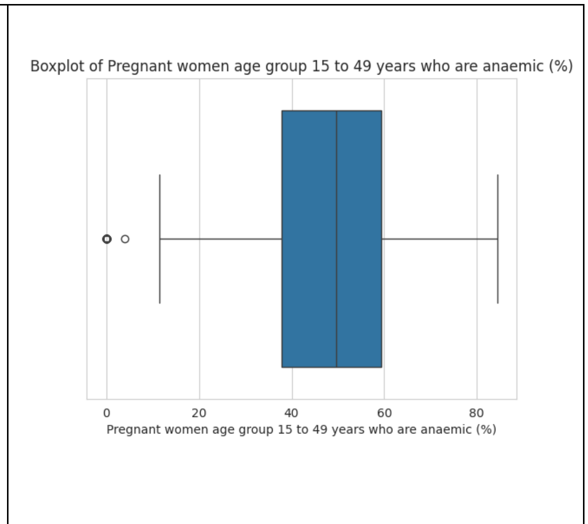
Insight:

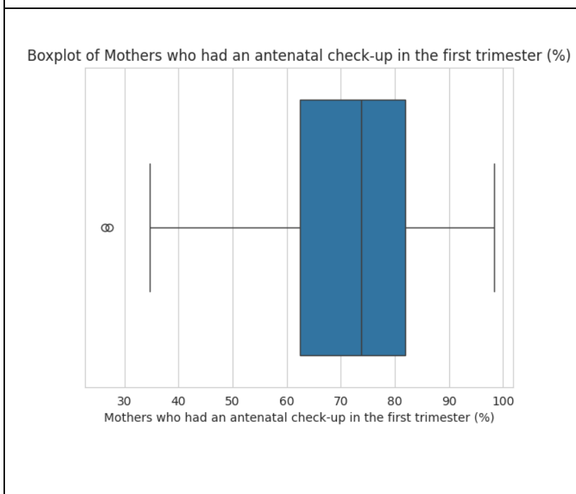
The distribution plots reveal that several healthcare and demographic variables are not symmetrically distributed, with noticeable skewness in variables such as ANC registration and institutional births. This suggests that a few districts with extreme values (often due to population size or regional advantages) dominate the distribution. Such skewness may influence model performance and indicates the need for scaling or transformation during modelling.

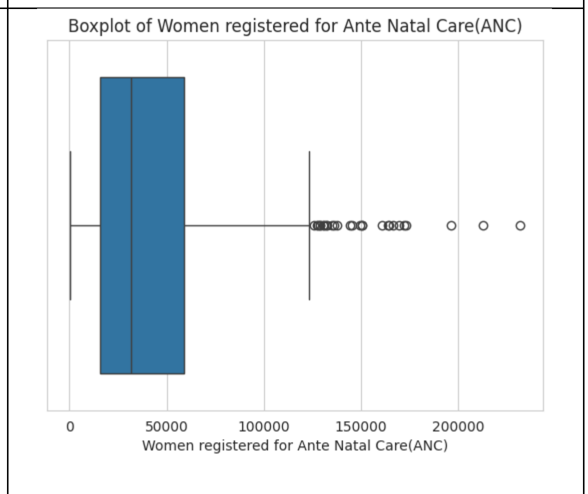
Additionally, variables related to maternal health and nutrition show broader distributions, reinforcing the presence of inequality in healthcare access and outcomes across districts.

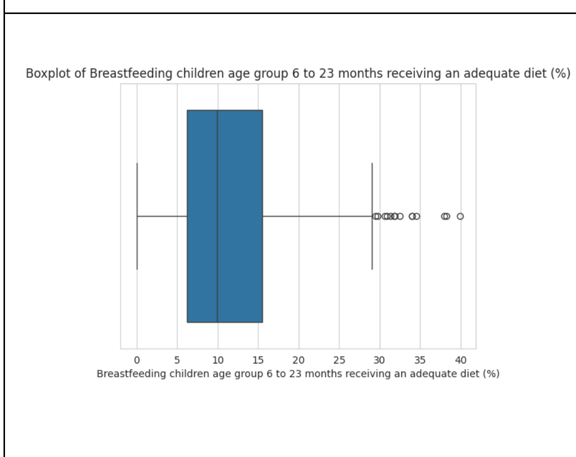
Table3: Univariate Analysis- Box plots

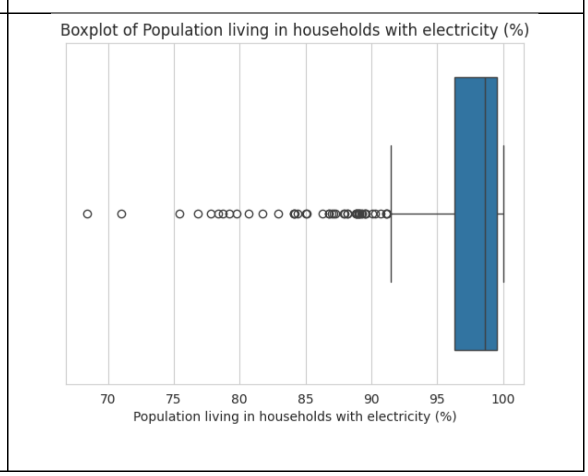












3. Univariate Analysis – Box Plots

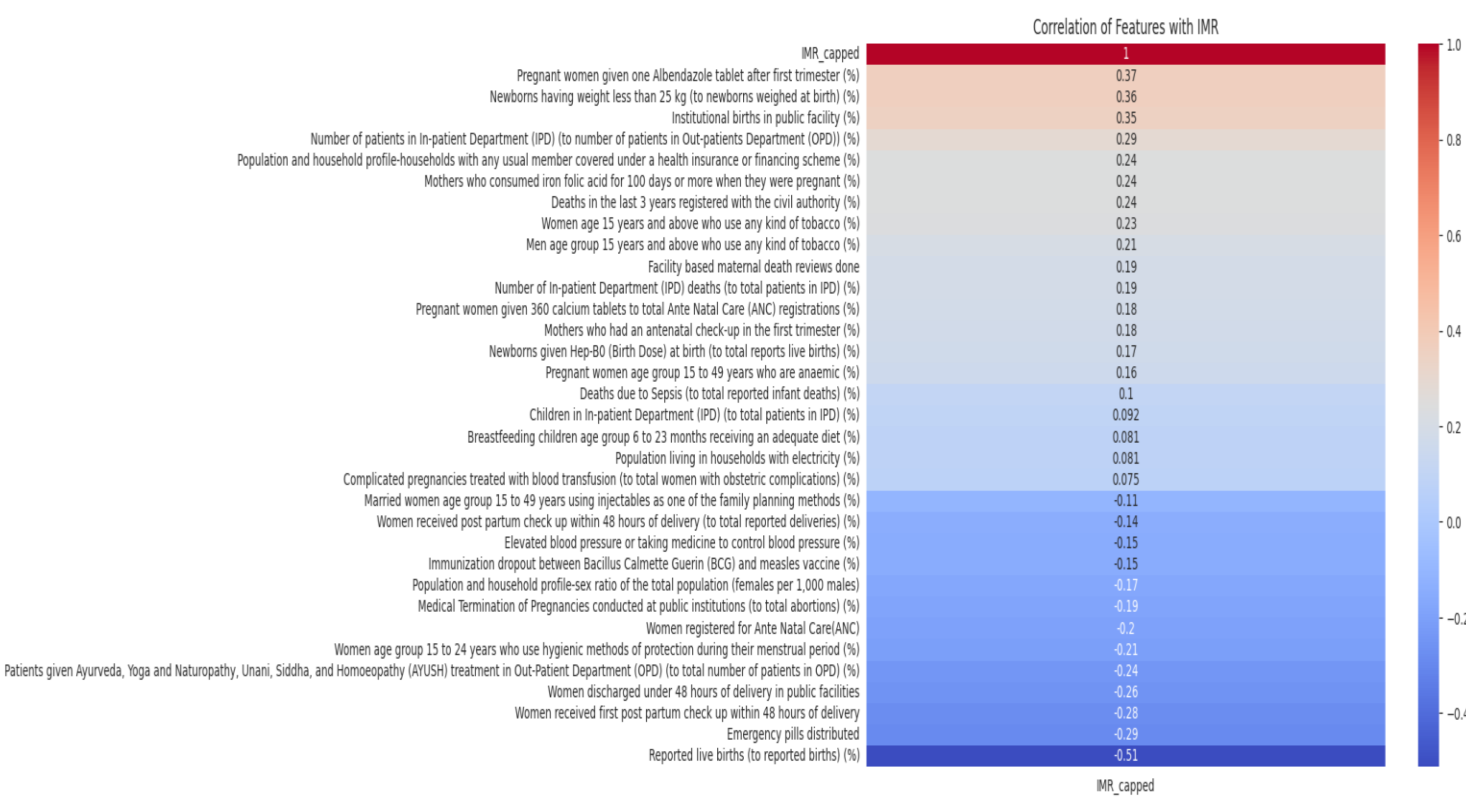
Box plots highlight the presence of **outliers and variability** across key features.

Insight:

The box plot analysis confirms the existence of multiple outliers across several variables, particularly in ANC registration and maternal health indicators. These outliers likely correspond to highly populated or better-resourced districts and can disproportionately influence statistical summaries and model estimates.

The variation in interquartile ranges across variables further emphasizes that some indicators (e.g., maternal anaemia, institutional births) have substantial spread, while others (e.g., electricity access) are tightly clustered. This suggests that not all features contribute equally to explaining variability in infant mortality.

Figure1: Bivariate: Correlation of Features with IMR



4. Bivariate Analysis – Correlation of Features with IMR

This analysis evaluates the **strength and direction of relationships** between predictors and infant mortality.

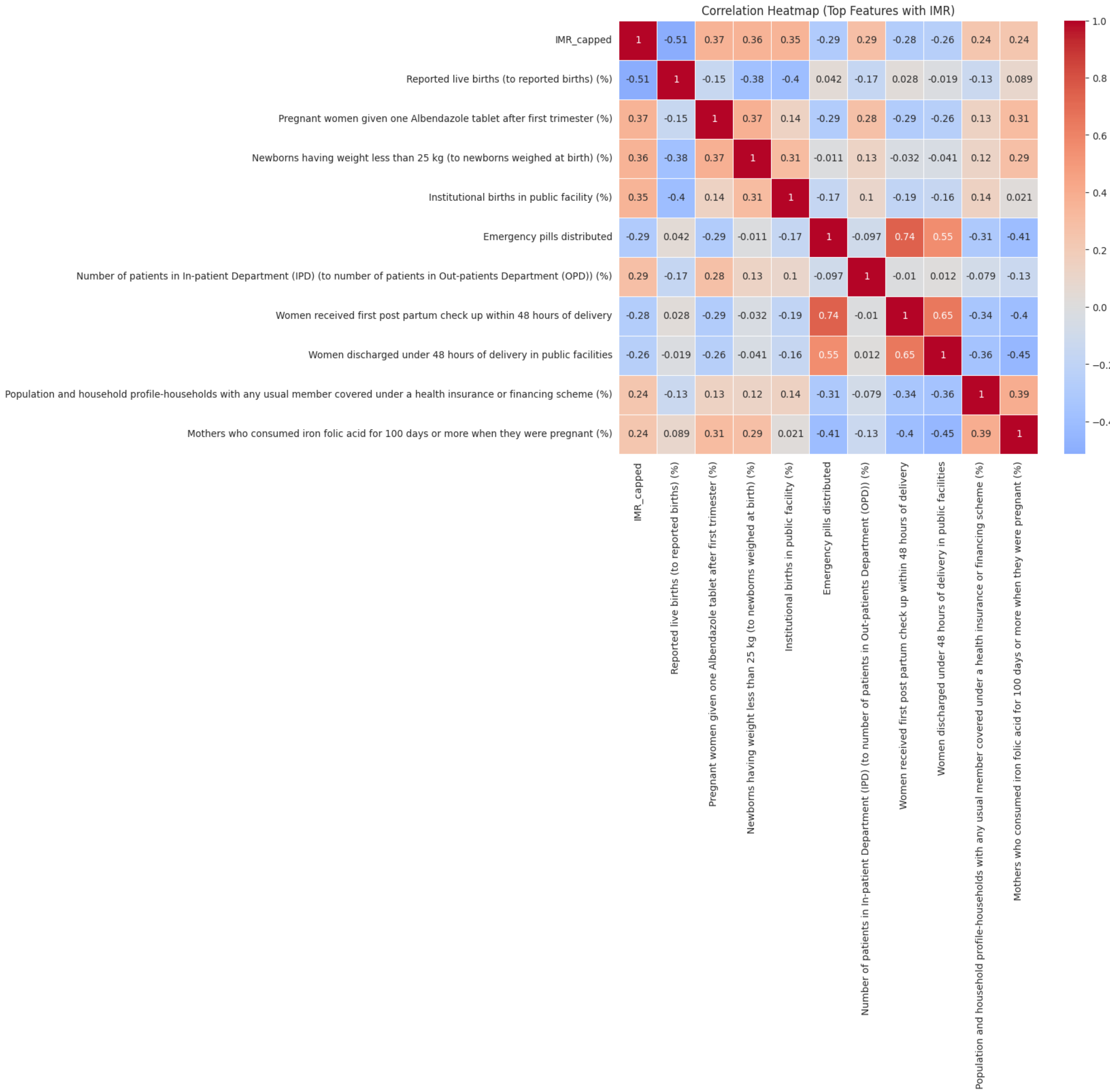
Insight:

The correlation analysis indicates that maternal health and healthcare access variables show stronger relationships with IMR compared to infrastructure variables. Features such as maternal anaemia and institutional births are expected to exhibit meaningful associations with IMR, as they directly influence maternal and neonatal health outcomes.

In contrast, variables with limited variability, such as electricity access, show weak correlations with IMR, suggesting minimal explanatory power despite their importance in general development.

Overall, this highlights that **health-specific indicators are more relevant predictors of infant mortality than general infrastructure indicators**.

Figure2: Bivariate: Correlation Heatmap (Top Features with IMR)



5. Bivariate Analysis – Correlation Heatmap (Top Features with IMR)

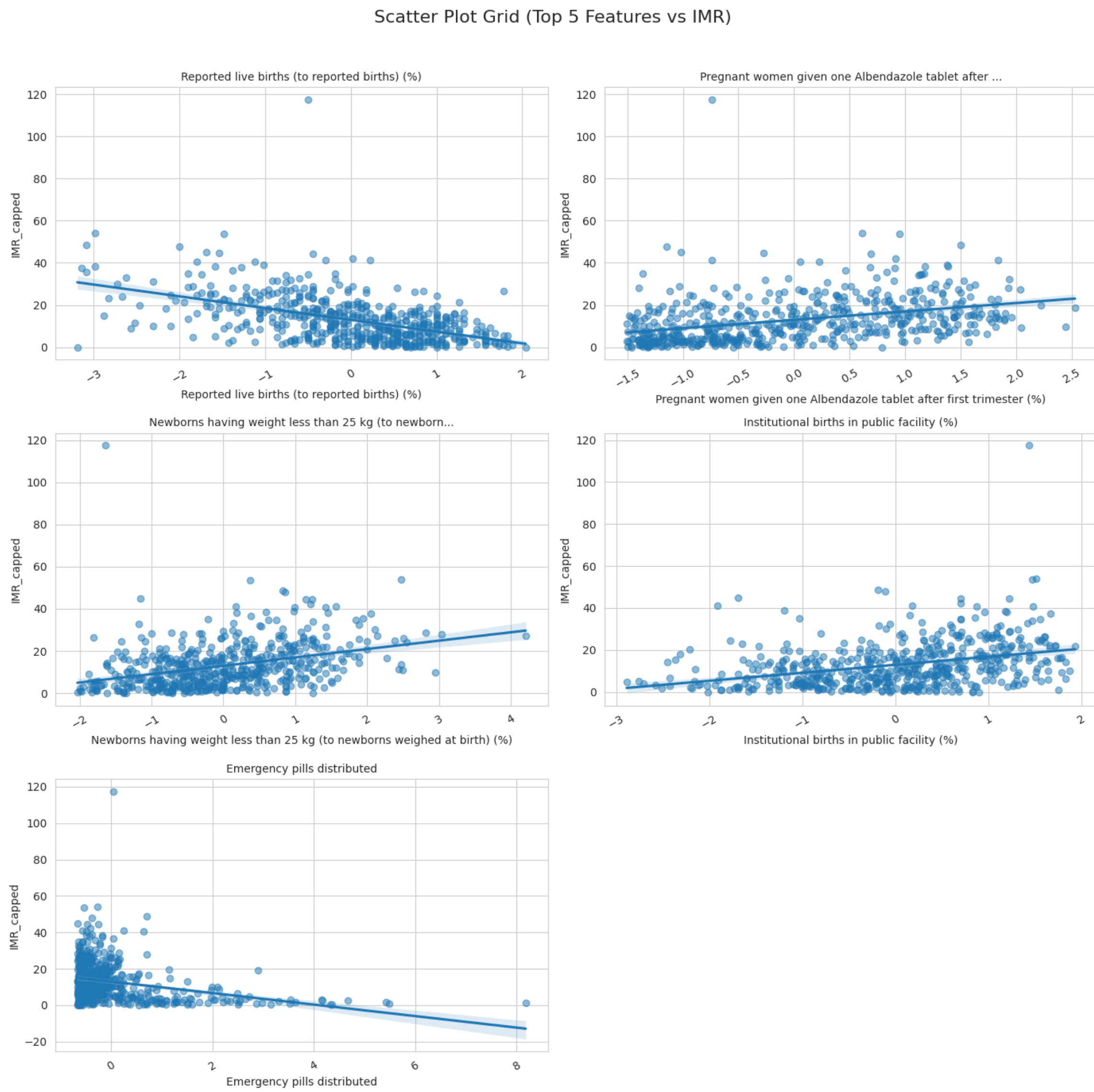
The heatmap provides a focused view of **key relationships among top predictors**.

Insight:

The correlation heatmap reveals clusters of related variables, indicating potential multicollinearity among maternal and healthcare indicators. For example, variables related to maternal health, nutrition, and healthcare utilization tend to be interrelated, reflecting underlying systemic factors such as healthcare access and socioeconomic conditions.

While these relationships strengthen their collective importance, they may pose challenges in modeling, particularly for linear models, where redundant information can affect coefficient stability. This suggests the need for feature selection or regularization techniques.

Figure3: Bivariate: Scatter Plot Grid (Top 5 Features vs IMR)



6. Bivariate Analysis – Scatter Plot Grid (Top Features vs IMR)

Scatter plots help visualize **functional relationships and trends** between predictors and IMR.

Insight:

The scatter plot analysis shows that relationships between predictors and IMR are generally **non-linear and dispersed**, rather than strongly linear. This indicates that changes in individual variables do not always result in proportional changes in IMR, suggesting the influence of multiple interacting factors.

Some variables may show weak but consistent trends, while others exhibit high variability with no clear pattern. This reinforces the idea that infant mortality is a complex outcome influenced by a combination of health, demographic, and socioeconomic factors.

The absence of strong linear patterns also suggests that **non-linear models or interaction effects may better capture these relationships**.

Summary

The univariate and bivariate analyses collectively highlight significant variability, skewness, and interdependence among key maternal and healthcare indicators. While certain variables demonstrate meaningful associations with infant mortality, the relationships are often complex and non-linear. The presence of outliers, skewed distributions, and multicollinearity underscores the importance of careful preprocessing and feature selection in subsequent modelling stages.

Figure1: Bivariate: Correlation of Features with IMR

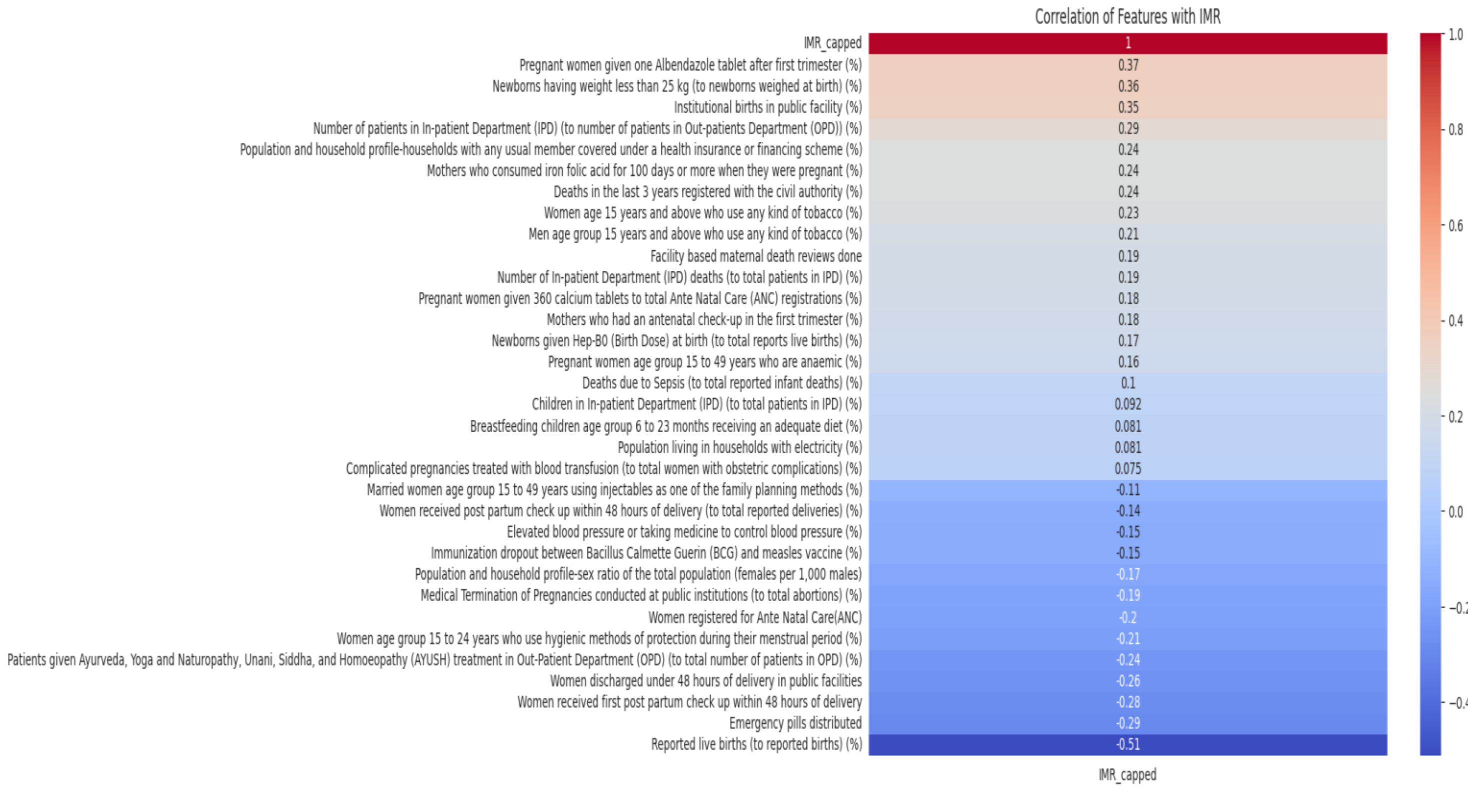


Figure2: Bivariate: Correlation Heatmap (Top Features with IMR)

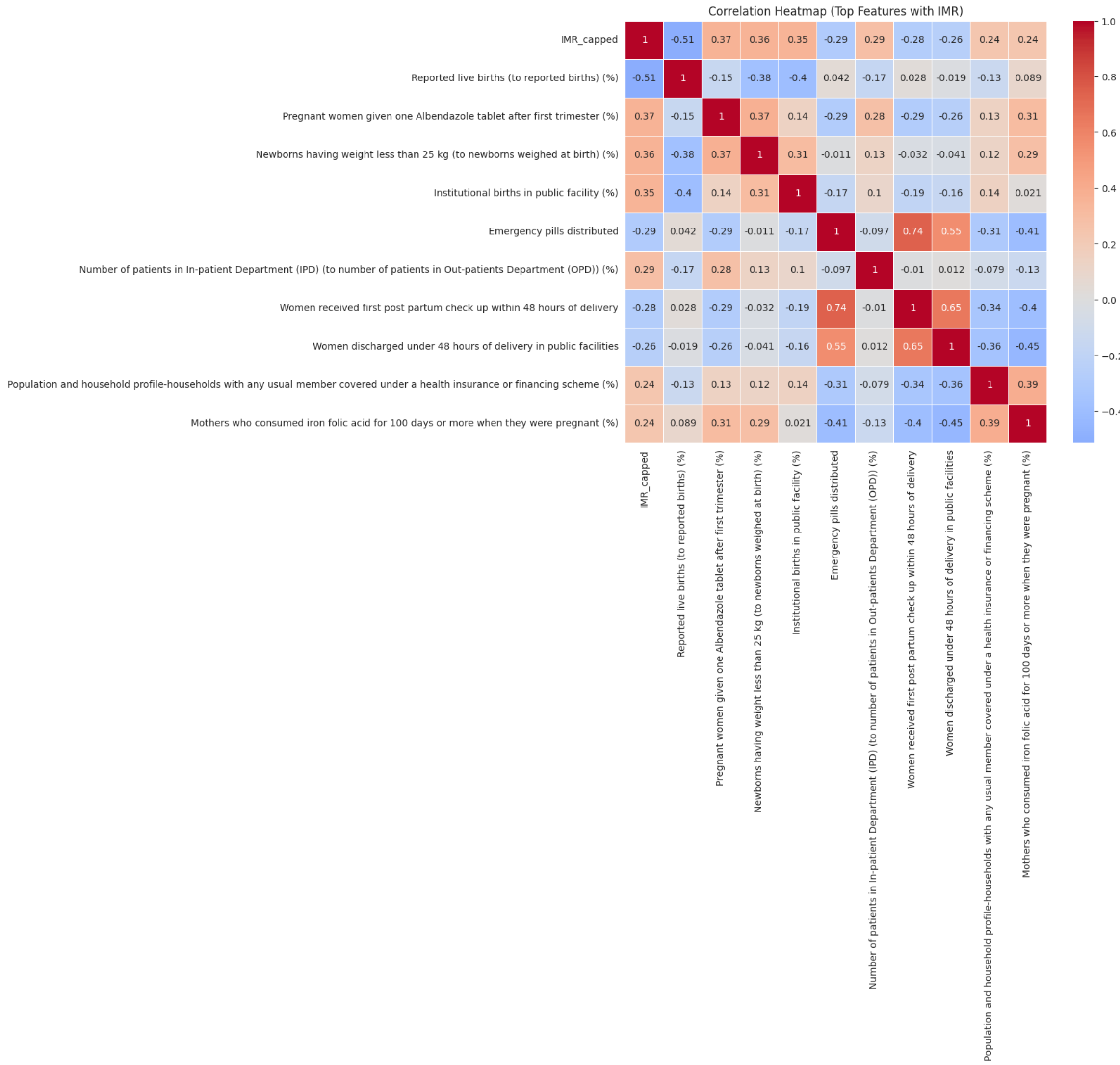


Figure3: **Bivariate: Scatter Plot Grid (Top 5 Features vs IMR)**